

**MINISTRY OF EDUCATION AND TRAINING
FPT UNIVERSITY**

**A LIGHTWEIGHT IMAGE COLORIZATION MODEL AND
APPLICATION IN COLORIZING LEGACY IMAGES OF
VIETNAMESE SOLDIERS**

by

Pham Van Thanh

A thesis submitted in conformity with the requirements
for the degree of Master of Software Engineering

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Assoc. Prof. Ph.D Phan Duy Hung

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Degree Master of Software Engineering

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Abstract

For the problem of grayscale image colorization, many authors propose their methods to produce the most plausible, vivid images from a gray input. Almost all of them introduce a quite large neural network model which consumes lots of time to colorize an image. This thesis proposes a relatively lightweight model to solve the problem which has equivalent performance to recent methods. Our model is based on famous U-net architecture which is frequently used for semantic segmentation problems. The model is trained to predict the chromatic ab channels given the lightness L channel in Lab color space to finally produce a colorful image. Experiments on commonly used PASCAL VOC 2012 and Places205 datasets show that our method has equivalent performance compared to other state-of-the-art algorithms while the model size is relatively smaller and consumes less computing resource. The model is also trained with images of Vietnamese modern soldiers. The trained model is used to build an application for coloring legacy images of Vietnamese soldiers during the Vietnam wars.

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1 Introduction to the problem of image colorization

1.1 Image colorization and applications

Image colorization is to assign a color to each pixel of a grayscale image. This is a difficult problem because two of the three channels of the ground truth image are lost. There only thing researchers can do is to find the most plausible color for each pixel instead of finding the lost original color. This is called semantic colorization.

Figure 1 shows a grayscale image (left) and its corresponding ground truth (right). In fact, the grayscale image is the result of a converting process where the ground truth image is the input. Colorization is the reverse process where researchers try to colorize the grayscale image and produce some colorful image like the ground truth. The colorization problem has multiple plausible solutions because of color diversity of objects. Therefore, colorizing process can never produce the same colors as the ground truth for every pixel. And the quality of a colorized image is usually evaluated on the closure to the ground truth as well as the plausibility and diversity of colors.



Figure 1. A grayscale image and its corresponding ground truth.

Colorization is an interesting area with many applications. One of the most popular applications of image colorization is to colorize legacy images or videos. This helps people to gain more knowledge about the colors of old images. A historical movie is more vivid when it is colorized and gives watchers a better imagination from what they see.

In Vietnam, there are many families who have their sons, fathers, or husbands sacrificed during the Vietnam wars. The only memory they left to their families could be a grayscale image of their portrait. Colorizing the grayscale images to make it vivid and colorful can be a way to appease their family members who live with the pain of losing their beloved ones. However, images of those soldiers can be damaged through a long time with no preservation in the right way. For such images, recovery is required before going to colorize.

Another thing to be considered in image colorization is that there are no ground truth images for the legacy images which was originally taken with old technology. Therefore, there are no training set with target output available for the legacy images. The best things researchers can do is to collect modern colorful images of the same topic for training and validation. And the trained model is used to colorize the legacy images with expectation that each pixel of the images is assigned with the most plausible and closest colors to the lost ground truth.

1.2 Related works

Before the era of deep learning, most of the colorization methods require human intervention. Previous works of [1] require color scribbles while [2], [3], [4] require reference images. During the emergence of deep learning, many authors like [5], [6], [7], [8], [10], [11] introduce their uses of deep convolutional neural networks (CNN) to solve this problem in a fully automatic approach while [9] proposes a deep learning method for user-guided image colorization. Their objective is to colorize the images with the most plausible colors instead of finding the lost original colors.

In deep learning approaches, authors proposed deep CNNs to predict the chromatic ab channels in Lab color space given the lightness L channel (Figure 2). The CNNs usually have a contracting path that contains downsampling steps and an expanding path containing upsampling steps to grow the image back to its original resolutions. Their models perform well and generate images that even can fool humans in a “colorization Turing test” where people are asked to choose which is a generated image or the ground truth image [5]. Most of the authors highlight the importance of semantic colorization. This is the key concept to colorize images plausibly.

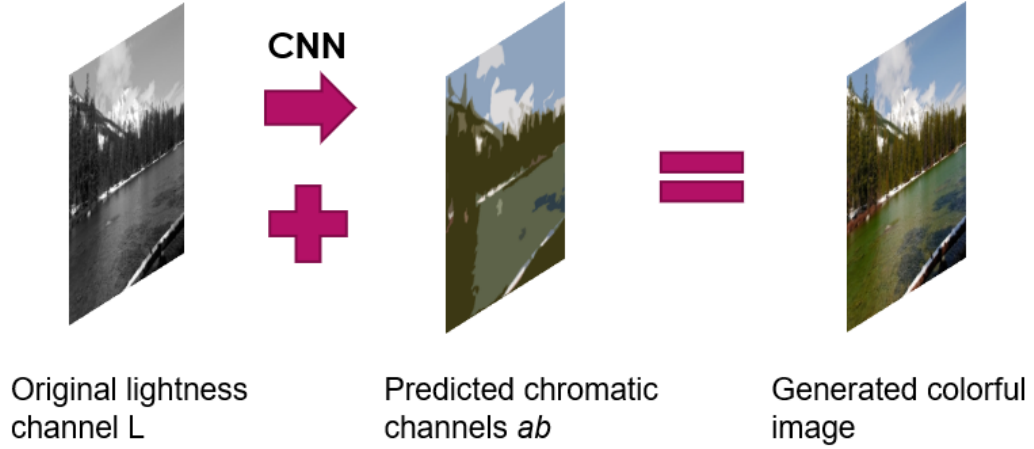


Figure 2. Deep learning approach of image colorization.

Furthermore, some authors like [8], [10], [12], [13], [14] focus on the diversity of the generated colors. A car can be red, yellow, or white, and a bird can be green, blue, or black. Their methods are supposed to predict more vivid images with diverse colors instead of assigning a single color for a class of objects.

To improve performance, other authors like [15] present a memory-augmented model that can produce high-quality coloration with limited data, or [16] use a pre-trained Inception-ResNetv2 to extract features of images in parallel with a normal encoder. [17] propose a method of instance-aware colorization. They leverage an off-the-shelf object detector to obtain object images and use an instance colorization network to extract object-level features.

1.3 Motivation and main contribution of this work

The above studies show that the problem of image colorization can be solved by different approaches. Many state-of-the-art algorithms are proposed to colorize images with the care of semantics and diversity. However, the uses of deep CNN models prevent people from deploying a colorization application on limited computational devices. A common property of the CNNs proposed by the mentioned authors is that they are quite large. The model of [5] has 129 MB of parameters, while [7] introduces a model which consists of 588 MB of parameters. Heavy models with too many parameters take a lot of time to colorize an image and much more time for a video or long movies. This motivates us to find a relatively lightweight model which has good

performance as most recent models and is light enough to be deployed and run smoothly on regular devices.

The main contribution of this study is to propose a lightweight model for semantic colorization which has equivalent performance to state-of-the-art colorization methods. It is supposed to predict the ab channel given the lightness L channel which can colorize images of any resolution in a fully automatic manner. The model is inspired by a successful semantic segmentation architecture called U-net proposed by [18].

This work applies self-supervised representation learning where raw images are used as the source of supervision. Each colorful (RGB) image in the training set is converted to Lab color space with lightness L channel and chromatic ab channels. The model is trained to predict ab channels given the lightness L channel. In other words, L channel is the input and the ab channels are target output. As the target output is a part of raw input and can be automatically extracted from the raw input without human effort, researchers called this approach “self-supervised representation learning”.

2 Methodology

This section of the thesis presents the proposed architecture and the loss function. The model is supposed to predict the ab channels from a given L channel in the Lab color space. There is a pre-processing step to convert each RGB image in the training set to a pair of input (L channel) and labeled output (ab channels). After prediction, the calculated ab channels are combined with the original lightness channel to produce the predicted Lab image. Images of any resolution are resized to 256x256 before processing and restored to the original resolutions after prediction.

2.1 The architecture

The U-net architecture is proposed by [18] which is based on the work of [19]. It is originally used for biomedical image segmentation. The CNN proposed by this study is a simplified version of the U-net (Figure 3). The U-net is proved to work well on semantic segmentation problem, and that is why it is applied to this model for semantic colorization which has some common characteristics to the segmentation problem. Like some other colorization methods, this network has a contracting path and an expanding path. However, skip connections are added to retain information that might be lost during the downsampling.

To make the model lightweight, the resolutions of input and other layer output are modified from the original U-net proposed by [18]. In the original U-net, the input resolution is 572x572 and output resolution is 388x388. For simplifying, the resolutions of both input and output of this model are 256x256. Furthermore, each convolutional block of the proposed architecture of this work has only one convolutional layer instead of two like in the original U-net. Another major modification is on the depth of activations. In origin U-net, the deepest activation has 1024 channels while the deepest one in the proposed model has only 512 channels.

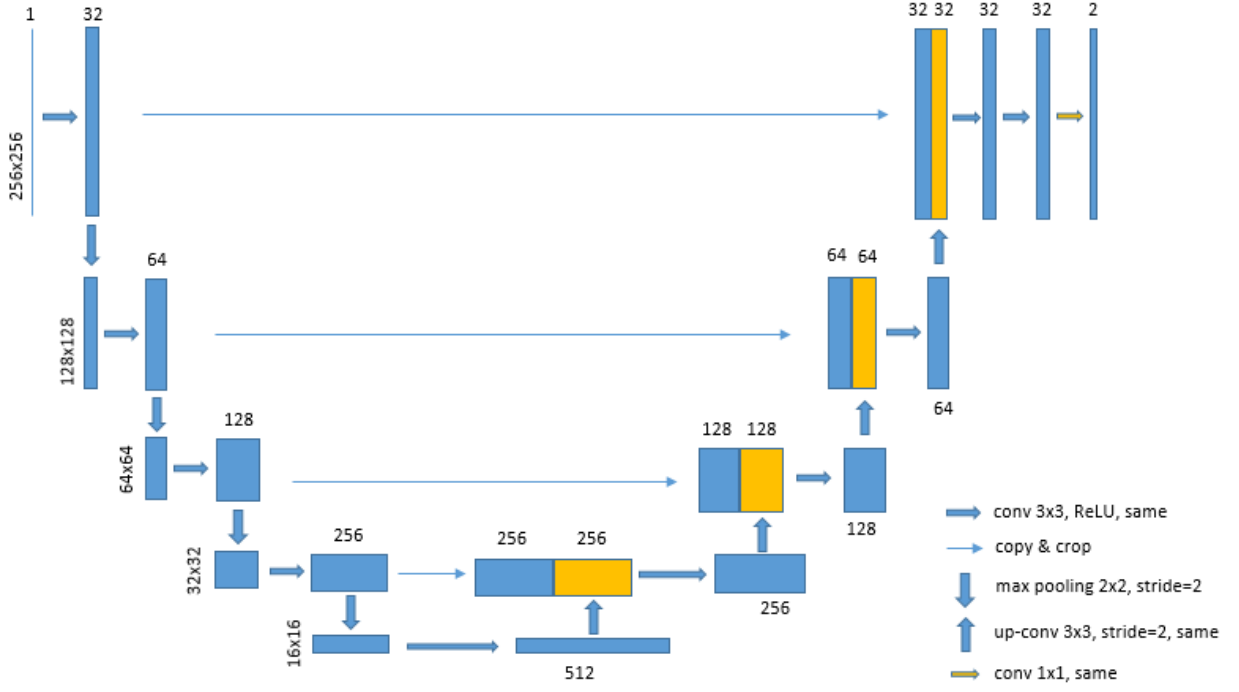


Figure 3. The simplified U-net architecture.

In Figure 3, the number of channels is denoted on the top of the box while the resolution is on the left side. Inputs are $256 \times 256 \times 1$ tensors where 1 represents the lightness channel L. Outputs are $256 \times 256 \times 2$ tensors representing 2 chromatic channels a & b. In the contracting path, 3×3 kernels and ReLU activations are used for all convolutional layers, and 2×2 max pooling for downsampling. Unlike [18], in this method, “same” paddings are applied in all convolutional layers.

In the expanding path, each upsampling step is done by an up-convolution that halves the number of channels and concatenation with a corresponding cropped feature map from the contracting path, followed by a 3×3 convolution with ReLU activation and “same” padding.

Finally, a 1×1 convolutional layer with 2 filters is used to predict the ab channels before they are combined with the original lightness L channel to produce a colorful Lab image. Figure 4 shows the proposed model where each rectangular box represents a convolutional block.

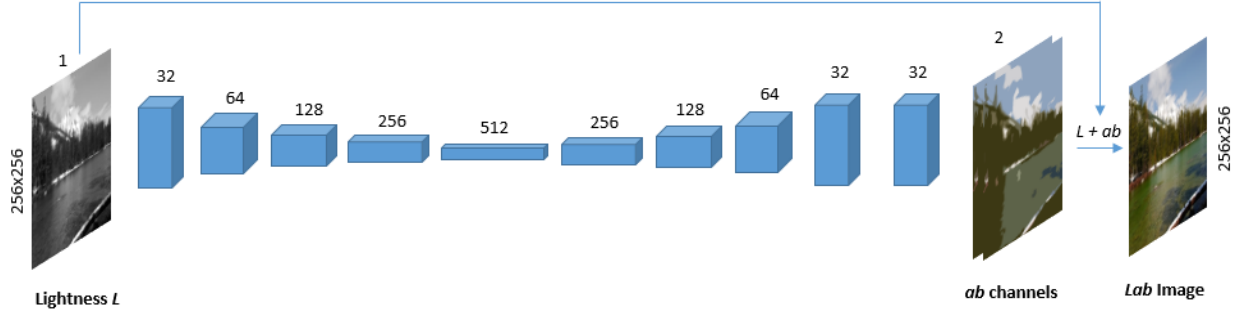


Figure 4. The proposed model.

This model consists of 9 convolutional blocks and a 1x1 convolutional layer. It has 4,711,714 parameters which are equivalent to 55 MB of memory storage. The model summary is shown in the following Table 1.

Table 1: Model summary.

Layer	Resolution	Channels	#Params
Input Layer	256	1	0
Conv2D	256	32	320
MaxPooling2D	128	32	0
Conv2D	128	64	18,496
MaxPooling2D	64	64	0
Conv2D	64	128	73,856
MaxPooling2D	32	128	0
Conv2D	32	256	295,168
MaxPooling2D	16	256	0
Conv2D	16	512	1,180,160
Conv2DTranspose	32	256	1,179,904
Concatenate	32	512	0
Conv2D	32	256	1,179,904
Conv2DTranspose	64	128	295,040
Concatenate	64	256	0
Conv2D	64	128	295,040
Conv2DTranspose	128	64	73,792

Concatenate	128	128	0
Conv2D	128	64	73,792
Conv2DTranspose	256	32	18,464
Concatenate	256	64	0
Conv2D	256	32	18,464
Conv2D	256	32	9,248
Conv2D	256	2	66

Compared to other models, this is relatively smaller and can be called “lightweight”. Table 2 shows the comparison where the model size of [5] is 2 times and [7] is 10 times larger than the proposed model approximately.

Table 2. Model size comparison of the proposed method and the others.

Method	Parameters (MB)
Zhang et al. 2016 [5]	129
Larsson et al. 2016 [7]	588
Proposed Model	55

For intuition, the three cubes in Figure 5 shows how lightweight the proposed model is when compared to models of Zhang and Larsson.

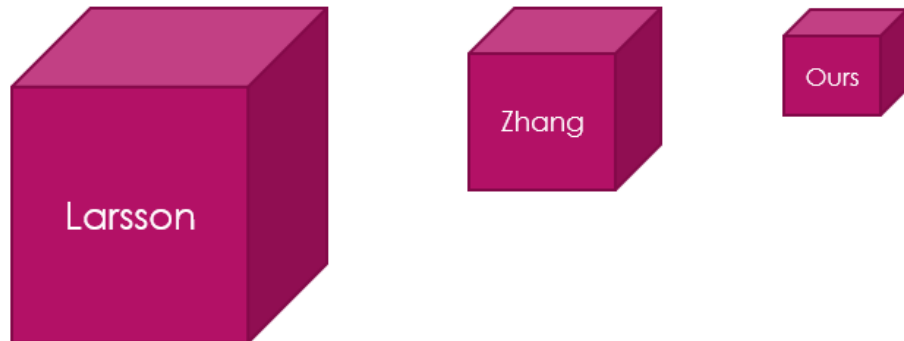


Figure 5. Intuitive comparison of the model sizes.

2.2 Loss function

Unlike [5] and some other authors who use custom loss functions, in this work, mean squared error (MSE) loss is used to evaluate the predicted mask. Mean squared errors is an estimator which measures the average of the squares of the errors. In this context, the error is the difference between the estimated value (predicted mask value) and the actual value (ground truth value). MSE is derived from the square of Euclidean distance, and its value is always greater than or equal to zero. The value of MSE decreases and approaches zero as the predicted mask get closed to the ground truth.

In statistics, the MSE is estimated by the following expression:

$$MSE = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2$$

Where N represents the number of entries, y_i represents the ground truth entry value and $\hat{y}_i$ represents the predicted entry value. In Lab color space, those values are in range of [-128, 127] for a & b channel.

For the problem of image colorization, researchers usually calculated the MSE over the chromatic ab channels (because the values of lightness L channel entries of the predicted mask and the ground truth image are exactly same). All the entries of a & b channels of the predicted mask are compared to the corresponding entries in the ground truth. And the MSE is the average squared errors of all the entry pairs.

Because the dimensions of predicted tensors are 256x256x2, the total number of entries in each tensor is:

$$N = 256 \times 256 \times 2 = 131072$$

And then the formula for calculating MSE in this problem is:

$$MSE = \frac{1}{131072} \sum_{i=1}^{131072} (y_i - \hat{y}_i)^2$$

As MSE is a standard loss function, it drives the model to learn and predict the mask that is as close as possible to the ground truth without considering much on vividity and colorfulness of the generated image. Sometimes, the predicted images can be grayish or desaturated because of this loss function. However, to make the model lightweight enough to build instant video colorization, a standard MSE is used instead of a complex loss function with much calculation and transformation.

2.3 Evaluation metrics

This work uses two metrics for evaluate quality of prediction tasks. The first is RMSE (Root Mean Squared Error). This is standard metric that is directly acquired from taking the square root of MSE defined in section 2.2.

$$RMSE = \sqrt{MSE}$$

RMSE (and as well as MSE) shows how the predicted mask and ground truth are close to each other. Smaller RMSE indicates better predicted masks that are closer to the ground truth.

The second metric is PSNR (Peak Signal-to-Noise Ratio). In signal processing, PSNR indicates the ratio between maximum possible power of the signal and the power of noise. In the context of image reconstruction, PSNR is a term for ration between the maximum range of color channels and the difference between the values of predicted mask and ground truth regarding to the channels.

In LAB color space, L channel value range is [0, 100] while ab channel value range is [-128, 127]. Then, to compare the predicted ab channels and the ground truth, the maximum range of color channels is 255 (the difference between 127 and -128). To calculate PSNR, the following ratio is computed:

$$ratio = \frac{255}{RMSE}$$

RMSE is estimated over all the entries in a & b channels of the predicted mask and ground truth. However, this ratio may have a wide dynamic range as it become too high when RMSE is close to 0. Then PSNR is expressed as a logarithmic quantity using decibel scale.

$$PSNR = 20 * \log_{10}(ratio) = 20 * \log_{10}\left(\frac{255}{RMSE}\right)$$

In other way, PSNR can be directly calculated from the MSE via the following expression:

$$PSNR = 10 * \log_{10}\left(\frac{255^2}{MSE}\right)$$

A higher PSNR indicates a better predicted mask. Indeed, as the predicted mask get closer to the ground truth, the RMSE becomes smaller and the PSNR get larger. In case the predicted mask and ground truth are identical, the RMSE is equal to 0 and PSNR is infinitely large. In fact, for image colorization task of this model, a predicted mask is rarely identical to the ground truth as there are 131,072 (256 x 256 x 2) entries to compare between them and the probability for all entry pairs are identical are extremely small.

3 Implementation and Evaluation

3.1 Experiment preparation

The proposed model is trained and evaluated on two different datasets. The first is a common semantic segmentation dataset named PASCAL VOC 2012. This dataset contains high resolution images of various topics in the real life. There are approximately 33,000 images in both train and test sets of this dataset. This work merges the train/validation set and the test set into a bigger one and then split into 2 new subsets for training (31,792 images) and validation (1,440 images). Evaluation on PASCAL VOC 2012 is used to compare the performance of the model to other methods where metrics are also reported on this commonly used dataset.

The second dataset is Places205 where 227,880 images are for training and 12,101 images for evaluation. This is a big dataset which contains images of 205 groups of places like hotels, hospitals, offices, pagodas, deserts, or art galleries... Dimension of every image of this dataset are 256x256, and no resize step is required before training. Training the model on Places205 is to report how well the model works on large datasets. Furthermore, the trained model on Places205 is also used for qualitative comparison to some state-of-the-art methods as they trained their models on such a large dataset with millions of images like ImageNet.

The model is built using the Tensorflow framework, this is a common framework with strong support for deep learning researchers. The Tensorflow version is 2.7.0. Training and evaluation processes are done with the Nvidia Geforce GTX 1650 GPU. The detailed information of this graphic card is shown in following Table 3.

Table 3. Nvidia Geforce GTX 1650 GPU Graphic Card Information.

CUDA Cores	896
Graphics boost clock	1155 MHz
Memory data rate	10.00 Gbps
Memory interface	128-bit
Memory bandwidth	160.03 GB/s
Total available graphics memory	8102 MB
Dedicated video memory	4096 MB GDDR6
Shared system memory	4006 MB

3.2 Model training

This work uses mini-batch training where the mini-batch size is 16. Adam optimization [20] is adopted. As the datasets are large, it is impossible to load an entire dataset to the training program for process. To resolve this problem, datasets are split into subsets. Each subset contains 240 images. This is to make sure the system does not encounter memory limit exceeding issue. An epoch is considered done when all the subsets of the dataset are loaded and processed.

When the model is trained on Places205, a single epoch contains 14,243 iterations and it averagely takes 7,500 seconds to complete. This includes the time of pre-processing the images before passing the tensors for model fitting. This is quite fast for such a large dataset like Places 205 which has more than 227,000 images in the training set. This fast learning enables researchers to make various changes to the model and quickly see the effectiveness during the tuning process.

The training loss is evaluated by calculating the average loss during each epoch while the validation loss is estimated after the entire epoch is completed. Therefore, in early epochs, the training loss is usually greater than validation loss.

As shown in Figure 6, the validation loss reaches the lower limit after 140,000 iterations (10 epochs). The lowest validation lost is 139.69 (RMSE = 11.82). Approximately, the training process takes 20 hours to get the best performance. After this point, the validation loss fluctuates between 140 and 150 while the train loss still decreases with a slower rate. This indicates that the hyperparameters of the model are well chosen and there are no problems like high-bias or high-variance. The detailed statistics of train loss and validation loss during training process are available on Appendix 1.

Places205 - Train Loss vs. Validation Loss



Figure 6. Train loss and validation loss on Places 205 Dataset.

On PASCAL VOC 2012, the model even learns faster. The Figure 7 shows how fast the learning progress is. It reaches the lowest validation loss after only 18,000 iterations. As each epoch is done in about 1,200 seconds, the whole training process takes only 3 hours to reach the best performance of the model.

Unlike when training on Places 205 dataset, in PASCAL VOC 2012, after the lowest point, the validation loss increases while the model still fits the training data better. This indicates model overfitting (high-variance) on such a small dataset as PASCAL VOC 2012. Because the lower limit of validation loss is an acceptable value, then validation-based early stopping [21] is adopted to terminate the training process when overfitting has begun. The statistics of train loss and validation loss during training process on PASCAL VOC 2012 are available on Appendix 2.

PASCAL VOC 2012 - Train Loss and Validation Loss

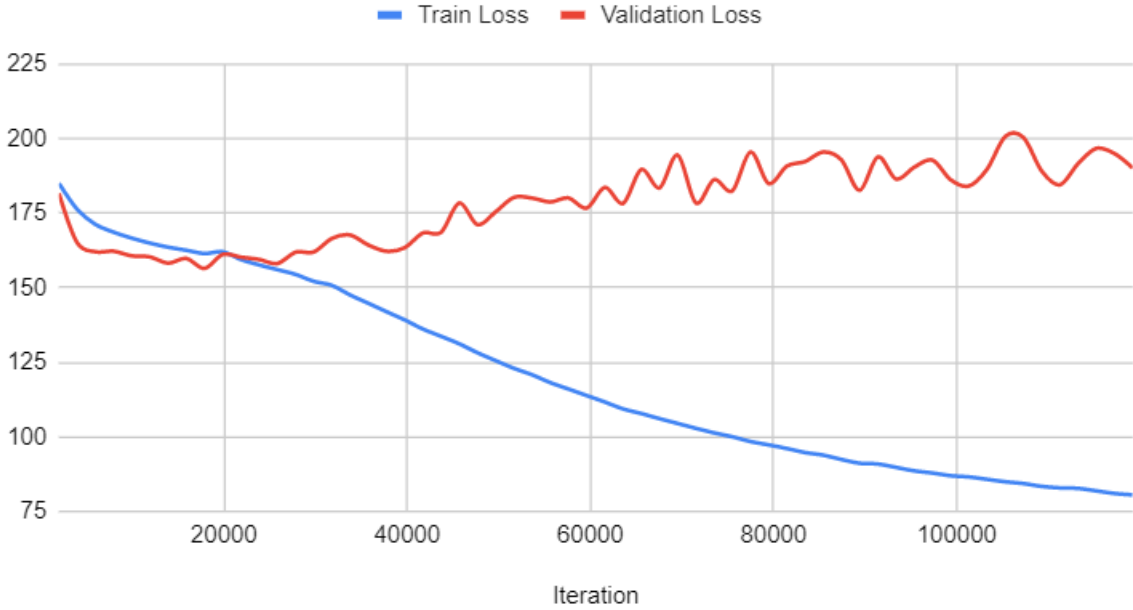


Figure 7. Train loss and validation loss on PASCAL VOC 2012 Dataset.

3.3 Quantitative comparison to other methods

Model performance is compared to some state-of-the-art colorization models proposed by [5], [6], [7], [9], and [10]. Two metrics used for evaluation are RMSE (Root Mean Squared Error) and PSNR (Peak Signal-to-Noise Ratio) measured on the predicted ab channels compared to the ground truth. While RMSE is a standard metric which directly measure how close the two tensors are, PSNR is a commonly used metric for evaluate quality of image reconstruction tasks.

The reports of [5], [6], [7], [10] are provided by the work of [10] while the metrics of [9] are reported by this study. Like [10], this work selects 1,440 images from Pascal VOC 2012 validation set to evaluate. The images are randomly selected from the set, and they could be different from the images selected by [10]. Therefore, the reported metrics of this work can be slightly different from the reports of [10] on the same method.

This study also reports the average prediction time of [5], [9], and the proposed model for speed comparison. This experiment is done on an Intel Core i7-10750H CPU to show how fast each

method runs on a regular device without powerful GPUs. The following Table 4 shows the experiment results reported on the proposed model compared to methods of [5] and [9].

Table 4. Quantitative evaluation to methods of [5] and [9].

Method	RMSE	PSNR	Runtime (ms)
Zhang et al. 2016 [5]	<i>12.80</i>	<i>23.49</i>	<i>349.65**</i>
Zhang et al. 2017 [9]	<i>10.52</i>	25.52	<i>1,182.15</i>
Proposed Model	<i>11.33</i>	<i>24.88</i>	112.98

Metrics in Table 4 show that the performance of proposed method is in the middle between the two methods while it is the fastest one in term of prediction time in a single image.

The reported metrics of this model is also compared to other methods of [6], [7], [10]. As mentioned before, the metrics of [5], [6], [7], [10] are provided by [10] while metrics of [9] are reported by this work (as the metrics of [9] are not available in the work of [10]).

Table 5. Quantitative evaluation to other methods.

Method	RMSE	PSNR	Runtime (ms)
Zhang et al. 2016 [5]	12.37	22.81	349.65**
Iizuka et al. 2016 [6]	10.66	24.20	-
Larsson et al. 2016 [7]	10.26*	24.56	-
Zhang et al. 2017 [9]	<i>10.52</i>	25.52	<i>1,182.15</i>
Zhao et al. 2020 [10]	11.43	23.15	-
Proposed Model	<i>11.33</i>	<i>24.88</i>	112.98

* Bold values indicate the best performance in comparisons.

** *Italic values indicate the metrics reported by this work.*

The results of Table 5 show that [7] outperforms the others on RMSE. The proposed method has a lower RMSE than [5], [10] and a higher one compared to [6], [7] and [9]. The method of [9] achieves the best performance regarding PSNR while the proposed method is at the second position. In summary, the proposed model performs better than most of the others according to the PSNR value and in the middle rank regarding RMSE. In terms of prediction time, this model is the fastest one as it is three times faster than [5] and ten times faster than [9]. The comparison

in prediction time to others method cannot be done due to the unavailability of the reported metrics.

3.4 Qualitative comparison

To show how good the generated images are, the proposed model is trained on Places205 and the predictions are compared to the ground truth images. Figure 8 shows how they “look like” each other. The images in Figure 8 are randomly selected from Places205 validation set. As mentioned before, this is a dataset of places where many of them are images of nature.

For some images like the right one in the first row of “the beach and sky”, or the left one in the second row of “the windmill”, the predicted mask is so close to the ground truth. The proposed method also works well on the left images of the third row of “the mini bar”. However, it does not predict a good result for the image of “the bedroom” in the third row, and the images of “the lake and forest” in the second row. For these two images, the predicted masks are quite grayish. In the image of the lake and forest, there is less green color and more blue color in the predicted mask than the ground truth. The trees are so grayish while the sky is still blue.

Experiments on other images of Places205 test set indicate that the model predicts better on blue areas like the sky or the ocean while it does not perform well on green objects like grass or trees. Generally, there is less green in the predicted masks than in the ground truth. This is called the “gray-effect”.

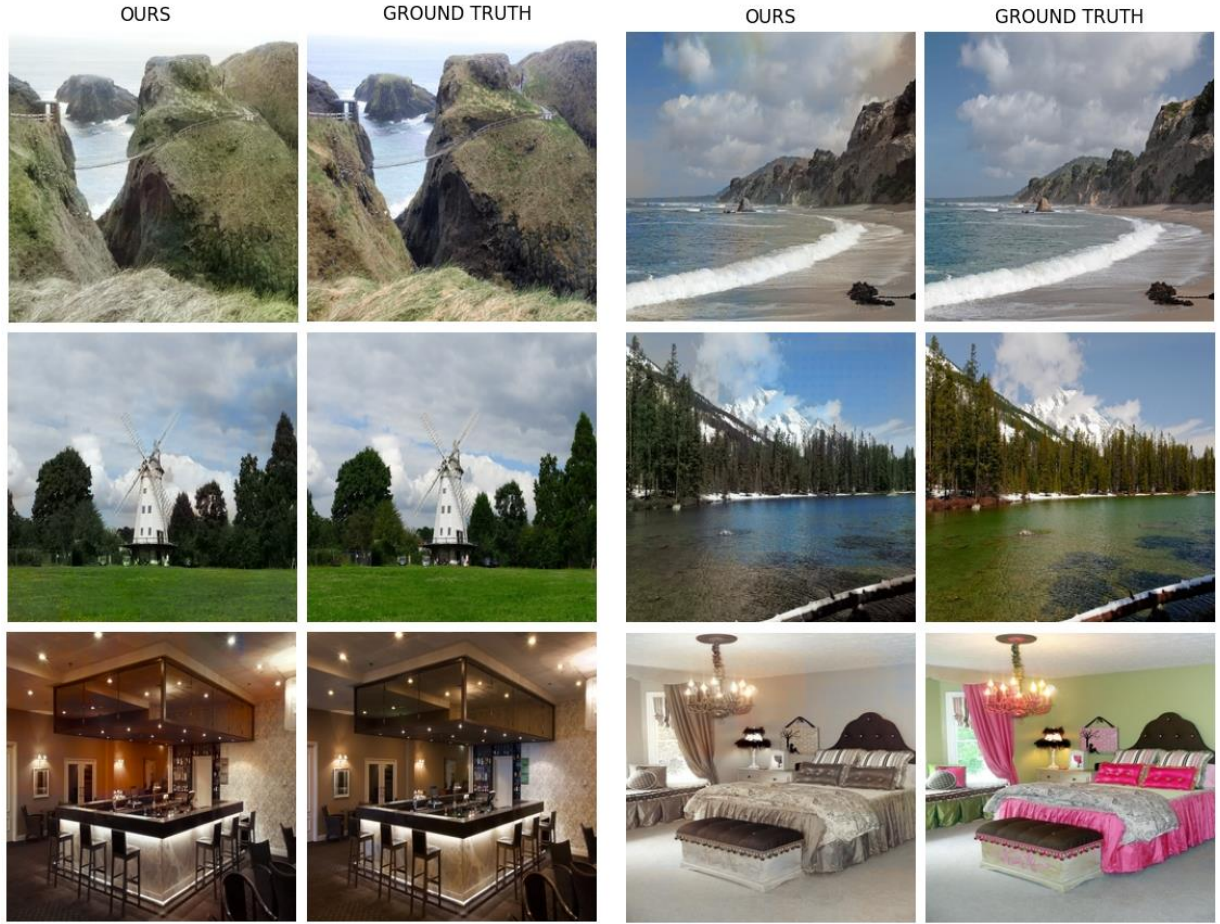


Figure 8. Predicted images of this work compared to the ground truths.

This work also compares the predicted images to the results of Zhang’s methods ([5] and [9]). These are two state-of-the-art models of semantic colorization. Figure 9 shows the comparison, from the left to right: Grayscale image; Prediction of [5]; [9]; Proposed Model and the Ground Truth.



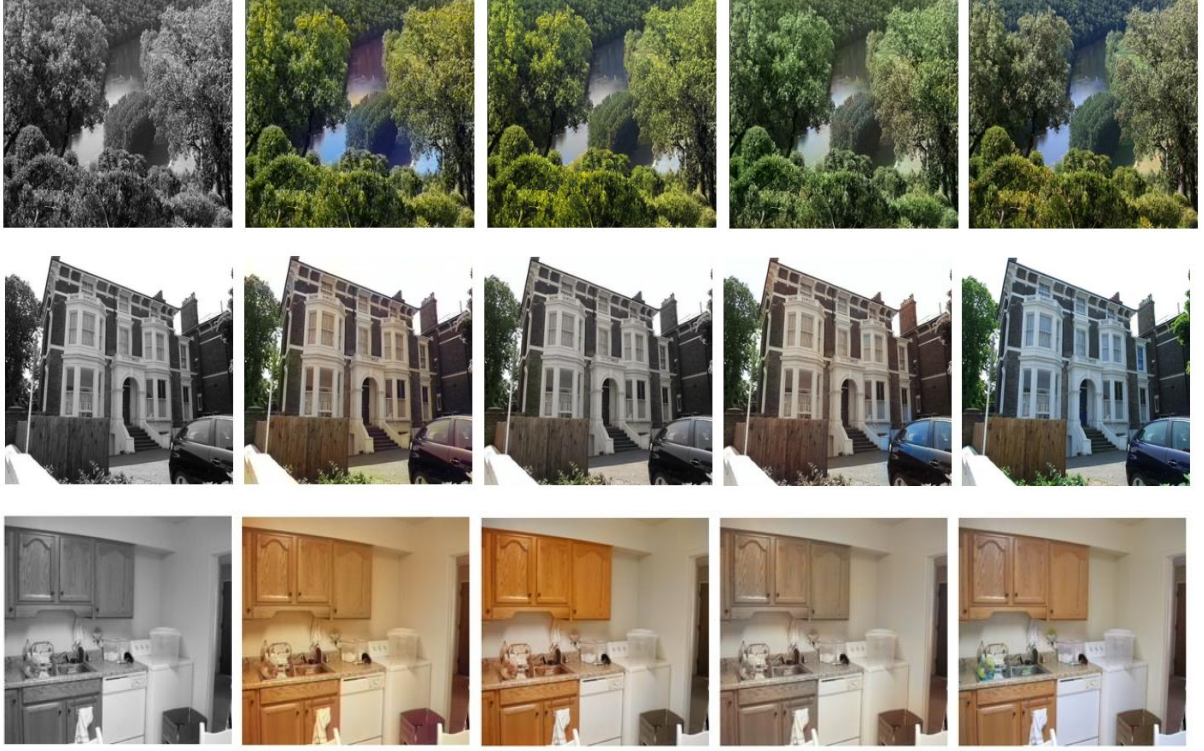


Figure 9. The prediction of the proposed method compared to Zhang’s and the ground truth.

Most of the images generated by [5] and [9] methods have more green color than the proposed method and even the ground truth. It makes their images more vivid and can fool people in a “colorization Turing test” [5]. According to [5], this is achieved by applying a customized loss function which is specially defined for the colorization problem. [5] also treats the colorization as a multinomial classification problem and build a model to learn a mapping to a probability distribution over possible color.

In the other hand, this “green effect” sometimes makes the images look unreal. For example, in the images on the first row of Figure 10, the railway in the prediction of [5] and [9] is green while it is not in the ground truth. And in fact, a railway is rarely green!

The other images in Figure 10 are also examples of the “green effect” of [5] and [9]. While the ground truth images are quite grayish, the predicted masks of [5] and [9] have more green color. This indicates that [5] and [9] methods are designed to make the predicted images as vivid as possible, no matter it is even more vivid than the ground truth.

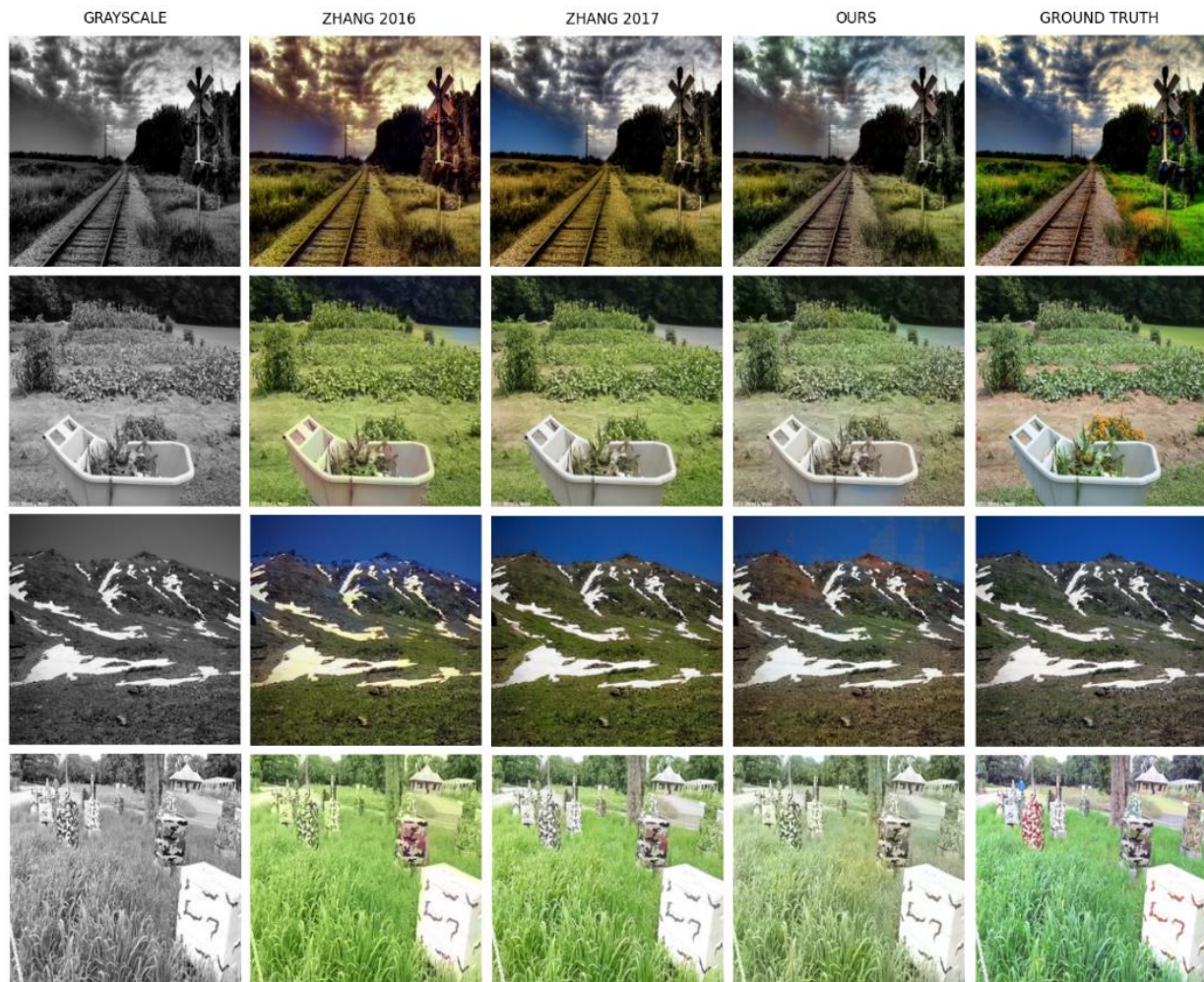


Figure 10. The “green-effect” in images generated by Zhang’s methods.

In general, the images generated by this work have the same closure level to ground truth as Zhang’s methods. However, because of the “green-effect”, Zhang’s methods do better on the images of nature where there is much green while the proposed model does better on the images of “less-green” things like cars, horses, lakes, and railways.

4 Colorizing legacy images of Vietnamese soldiers

Colorizing legacy images of Vietnamese soldiers which were taken during the Vietnam wars is a meaningful task as it could give their families a more vivid image of their sons, fathers or husbands. Most of the images taken for the troops during Vietnam wars are grayscale images due to the poor condition those times. Some of the images were damaged seriously so it takes a huge effort for recovery and colorization. As this thesis focuses on the task of colorization, only the grayscale images which were not damaged are chosen for evaluation.

4.1 Dataset preparation

A different dataset of modern Vietnamese troops is prepared for training. 347 portrait images of troops and 527 published images of Vietnamese army (including the troops and landscapes) are collected and preprocessed before training. The portrait images only include the faces, necks, shoulders and a part of the chest. Other parts of the body (if any) must be truncated from these portrait images.

Despite the portrait images, the landscape images include troops as well as the background. It can be an image of a march where the troops are moving on a path in a tropical forest, or on a hill, a mountain or rice field. These images are included to the trainset to get the trained model to work better on legacy images of troops that may have a background.

Both portrait images and landscape images are mixed into a set and shuffled before being split into two different sets. The trainset has 792 images and the validation set has 82 images.

An additional test set is prepared. That is a set of legacy images of Vietnamese soldiers. Only undamaged images are selected. The set contains 74 images including 19 portrait images and 55 general images.

4.2 Training

Mini-batch training is used with Adam optimization. Like the previous training on Places 205 and PASCAL VOC 2012, the mini-batch size is 16. However, in this small training set, the training process is much faster. Approximately, each epoch takes only 40 seconds to be completed. The Figure 11 shows how fast it is.

Soldiers Dataset - Train Loss vs. Validation Loss



Figure 11. Train loss and validation loss on Soldiers Dataset.

The validation loss reaches the lower limit after 5,000 iterations (100 epochs) that takes about 66 minutes. The lowest validation loss is 187.81 (RMSE = 13.70). After 100 epochs, the train loss continues decreasing while the validation loss slightly fluctuates around the point of 200. Unlike the training process on PASCAL VOC 2012, the model does not encounter overfitting issue when being trained on this dataset. The statistics of train loss and validation loss during training process on this dataset are available on Appendix 3.

4.3 Evaluation

The trained model (after 100 epochs or 5,000 iterations) has the best performance on validation set. And this is chosen to evaluate against the test set of legacy images of soldiers. Figure 12 shows some examples of colorization by this model. The left column represents grayscale images while the middle column is the predicted images by the proposed model. The ground truth images are in the right column.

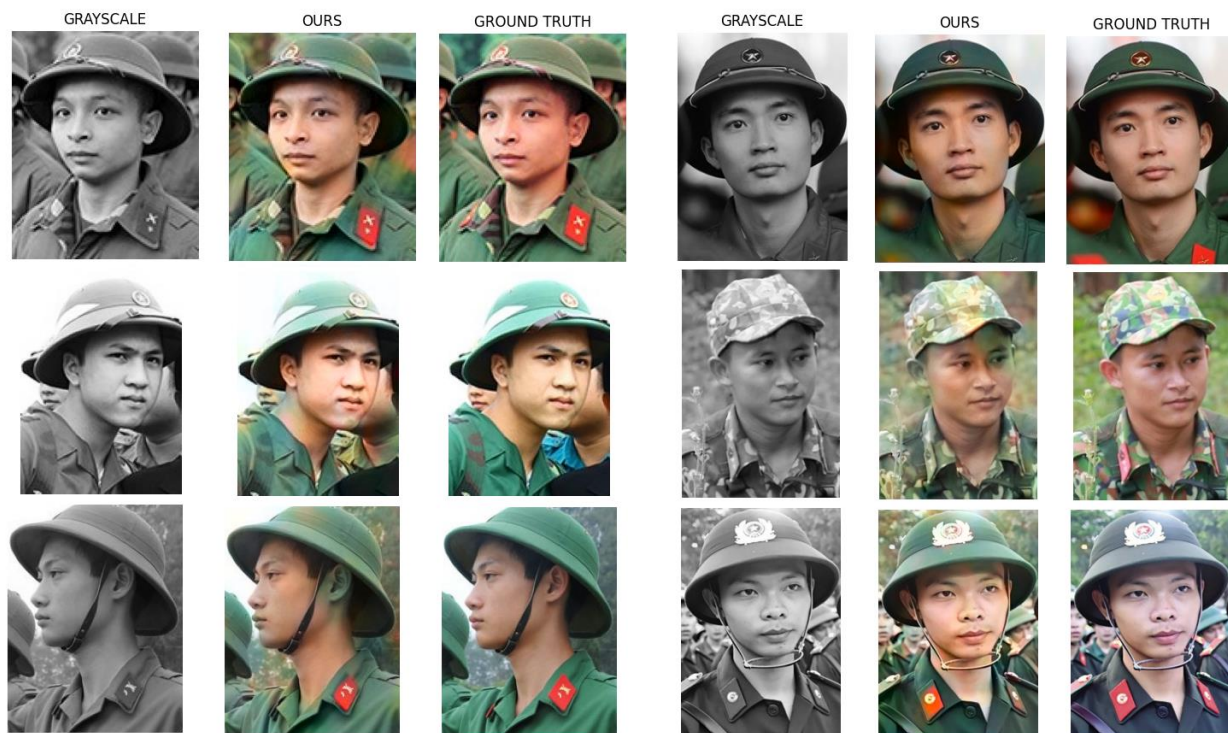


Figure 12. Soldier portrait images predicted by this model compared to the ground truth.

The images used for predictions in Figure 12 are selected from validation set. They are portraits of modern Vietnamese troops and the ground truth images are available for comparison. Generally, the predicted images are quite close to the ground truth. However, some predicted images are a little “desaturated”.

The trained model also colorized some legacy images as shown in Figure 13. The images of legendary General Vo Nguyen Giap of Vietnamese People’s Army and some other soldiers are colorized. As expected, the colorized images are much more vivid than the grayscale images and they give the viewers a better imagination of the soldiers even with a low-quality grayscale image.



Figure 13. Colorized legacy images of legendary General Vo Nguyen Giap and other soldiers.

4.4 Conclusion & Future works

This thesis proposed a neural network architecture for fully automatic image colorization. The model is a simplified version of U-net architecture which is relatively lightweight compared to other authors. The predicted images are not as vivid as some state-of-the-art methods like [5] and [9]. However, the images reach the ground truth as close as those do. The model size is several times smaller and then it runs fast even on a device with no GPU available. Therefore, it can be used to build instant video colorization applications where so many image frames need to be colorized in an acceptable time.

The proposed model is also trained with images of modern Vietnamese soldiers and the trained model is used to colorize legacy images of soldiers in the Vietnam wars. As the legacy images quality is usually poor, the colorized images are not so vivid as the images of modern troops. However, the quality of colorized images is acceptable and the trained model can be used for building a general application for colorizing legacy grayscale image.

Predicted images by this work are quite grayish and sometimes desaturated. It is due to the use of a standard loss function like MSE. In the future, a custom loss function is applied to the model for training and predicting more colorful and saturated images.

The quality of colorized images of the model can also be improved by collecting more modern images of Vietnamese troops for the training set. The current set has only 792 images for training and the trained model on such a small set does not generalize well.

After being trained in a larger dataset, the model can be used to build an application for legacy image and video colorization. As the model is lightweight and can run fast on non-GPU devices, the application can be deployed on mobile devices for instant video colorization. Users may view colorized legacy videos or movies for better experience.

Furthermore, the proposed architecture of this work can be used as the convolutional part of complicated colorization models. This architecture is lightweight while other complicated models improve colorization performance with various techniques are adopted. The use of proposed architecture on those models will create a lightweight model with better performance of colorization and will generate high quality images or videos in a fast manner. Finally, this work can be a good reference for many machine learning problems [22-24].

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Appendices

Appendix 1. Train loss and validation loss during training on Places 205.

Epoch	Iteration	Train Loss	Validation Loss
1	14243	167.07	156.82
2	28486	153.91	150.12
3	42729	149.06	146.53
4	56972	145.88	144.85
5	71215	143.37	145.41
6	85458	141.26	142.92
7	99701	139.37	142.32
8	113944	137.75	142.72
9	128187	136.27	143.44
10	142430	134.60	142.87
11	156673	133.19	139.88
12	170916	131.59	139.69
13	185159	130.19	140.55
14	199402	128.85	142.06
15	213645	127.45	142.75
16	227888	126.11	143.05
17	242131	124.83	143.04
18	256374	123.57	145.81
19	270617	122.32	147.18
20	284860	121.29	147.87
21	299103	120.17	142.68
22	313346	119.23	145.75
23	327589	118.24	145.87
24	341832	117.20	148.85
25	356075	116.48	145.13
26	370318	115.55	146.97
27	384561	114.76	146.62

28	398804	114.00	147.26
29	413047	113.27	146.94
30	427290	112.69	147.58
31	441533	112.06	145.68
32	455776	112.16	152.16
33	470019	111.04	147.29
34	484262	110.29	148.74
35	498505	109.79	148.99
36	512748	109.17	148.95
37	526991	108.73	147.39
38	541234	108.31	149.53
39	555477	107.87	148.65
40	569720	107.49	147.70

Appendix 2. Train loss and validation loss during training on PASCAL VOC 2012.

Epoch	Iteration	Train Loss	Validation Loss
1	1987	184.99	181.60
2	3974	176.28	165.09
3	5961	171.21	162.04
4	7948	168.61	162.21
5	9935	166.61	160.66
6	11922	164.93	160.31
7	13909	163.57	158.21
8	15896	162.54	159.80
9	17883	161.42	156.44
10	19870	162.04	161.13
11	21857	159.42	160.22
12	23844	157.62	159.45
13	25831	156.04	158.09
14	27818	154.45	161.87
15	29805	152.13	161.97
16	31792	150.76	166.51
17	33779	147.54	167.64
18	35766	144.76	164.30
19	37753	141.94	162.22
20	39740	139.18	163.39

21	41727	136.11	168.32
22	43714	133.68	168.57
23	45701	131.17	178.39
24	47688	128.17	171.11
25	49675	125.53	175.44
26	51662	122.95	180.26
27	53649	120.79	180.02
28	55636	118.16	178.73
29	57623	116.00	180.08
30	59610	113.83	176.67
31	61597	111.65	183.62
32	63584	109.34	178.23
33	65571	107.79	189.64
34	67558	106.08	183.42
35	69545	104.47	194.46
36	71532	102.84	178.39
37	73519	101.32	186.22
38	75506	100.00	182.43
39	77493	98.41	195.49
40	79480	97.29	184.88
41	81467	96.07	190.77
42	83454	94.72	192.28
43	85441	93.84	195.47
44	87428	92.39	192.81
45	89415	91.12	182.63
46	91402	90.88	193.85
47	93389	89.74	186.41
48	95376	88.58	190.39
49	97363	87.86	192.77
50	99350	86.95	186.16
51	101337	86.55	184.08
52	103324	85.73	189.71
53	105311	84.92	200.69
54	107298	84.33	200.33
55	109285	83.38	189.10
56	111272	82.87	184.49
57	113259	82.71	191.59
58	115246	81.87	196.68
59	117233	81.01	194.95
60	119220	80.53	190.12

61	121207	80.24	191.72
62	123194	79.32	202.40
63	125181	79.10	195.64
64	127168	78.55	193.25
65	129155	78.17	193.69
66	131142	77.95	193.87
67	133129	77.39	200.75
68	135116	76.73	194.21
69	137103	76.38	206.28
70	139090	76.19	197.16
71	141077	75.38	195.15
72	143064	75.10	197.18
73	145051	74.51	194.26
74	147038	74.51	194.59
75	149025	74.26	196.97
76	151012	73.73	188.17
77	152999	73.50	196.17
78	154986	72.92	196.96
79	156973	72.69	191.11
80	158960	72.44	192.57
81	160947	72.21	191.09
82	162934	71.86	200.48
83	164921	71.70	195.17
84	166908	71.55	194.79
85	168895	71.18	203.76
86	170882	70.78	191.47
87	172869	70.75	199.39
88	174856	70.31	193.89
89	176843	70.33	182.99
90	178830	69.83	195.81
91	180817	69.78	203.30
92	182804	69.49	195.83
93	184791	69.30	195.80
94	186778	69.40	198.54
95	188765	68.91	196.02
96	190752	68.36	194.95
97	192739	68.36	204.23
98	194726	68.20	208.02
99	196713	68.26	200.73
100	198700	67.75	200.17

Appendix 3. Train loss and validation loss during training on Soldier Dataset.

Epoch	Iteration	Train Loss	Validation Loss
5	250	258.90	258.22
10	500	254.87	256.92
15	750	241.48	266.46
20	1000	244.86	247.15
25	1250	243.31	263.01
30	1500	250.91	252.82
35	1750	251.62	251.09
40	2000	233.90	248.18
45	2250	216.09	214.37
50	2500	200.45	210.22
55	2750	174.50	199.99
60	3000	140.30	190.11
65	3250	117.64	198.71
70	3500	95.25	196.84
75	3750	85.71	191.22
80	4000	72.09	193.23
85	4250	66.03	198.30
90	4500	51.52	191.39
95	4750	45.31	188.19
100	5000	40.62	187.71
105	5250	40.59	193.50
110	5500	35.03	197.41
115	5750	36.30	205.33
120	6000	32.25	199.21
125	6250	31.98	195.30
130	6500	31.33	196.65
135	6750	28.82	196.39
140	7000	27.62	194.16
145	7250	26.97	196.26
150	7500	26.32	196.97
155	7750	31.03	199.01
160	8000	25.19	192.50
165	8250	23.39	195.98
170	8500	24.45	196.05
175	8750	23.93	200.95
180	9000	25.29	194.55
185	9250	20.33	197.45
190	9500	19.24	192.95

195	9750	19.50	194.22
200	10000	19.74	196.97
205	10250	18.77	192.01
210	10500	21.16	195.48
215	10750	20.32	197.95
220	11000	18.38	196.09
225	11250	18.82	197.38
230	11500	16.28	190.66
235	11750	22.03	193.13
240	12000	15.82	193.06
245	12250	15.20	188.82
250	12500	14.69	188.66
255	12750	15.79	190.19
260	13000	15.76	193.40
265	13250	15.40	191.45
270	13500	15.91	192.63
275	13750	14.73	188.81
280	14000	13.96	191.19
285	14250	15.92	191.90
290	14500	15.74	189.90
295	14750	16.30	192.20
300	15000	17.18	193.28
305	15250	13.88	193.17
310	15500	13.45	190.55
315	15750	14.67	190.86
320	16000	14.15	191.06
325	16250	14.14	192.78
330	16500	13.50	190.25
335	16750	13.03	191.85
340	17000	12.44	188.87
345	17250	12.52	188.80
350	17500	13.39	189.96
355	17750	12.86	188.20
360	18000	11.76	188.79
365	18250	12.79	189.12
370	18500	12.38	188.70
375	18750	12.09	188.56
380	19000	13.09	188.26
385	19250	13.08	192.09
390	19500	13.87	192.81

395	19750	13.08	191.31
400	20000	13.31	194.10
405	20250	10.80	188.46
410	20500	10.97	189.40
415	20750	10.70	189.36
420	21000	10.37	187.96
425	21250	10.75	189.41
430	21500	11.53	188.59
435	21750	11.59	190.18
440	22000	11.69	192.66
445	22250	12.49	191.16
450	22500	12.08	190.18
455	22750	11.15	192.03
460	23000	11.72	193.38
465	23250	11.14	191.14
470	23500	11.04	192.53
475	23750	11.06	192.84
480	24000	10.35	191.27
485	24250	10.65	189.95
490	24500	9.70	189.84
495	24750	9.93	189.07
500	25000	10.36	191.04